



Circular Knitting

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Background

1.1 Context

This document summarises the modelling approach for **circular knitting**, representing a typical industrial process for jersey fabric production.

The purpose is to make the process transparent and reproducible across studies, while allowing refinement as factory-specific data become available.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of knitted fabric.
- **Technology:** single and double jersey circular knitting.
- **Included:** yarn losses/waste, lubricants, energy consumption, electricity, indirect energy allocation, capital goods and their end-of-life (EoL).
- **Excluded:** fibre and yarn manufacturing (modelled separately), building construction, packaging and logistics.

Methodological Overview

2.1 Yarn losses and lubricant - dependency on yarn fineness (dtex)

Concept

Circular knitting waste and lubricant use vary with yarn fineness.

Finer yarns require longer machine time, experience higher friction, and tend to break more often — leading to higher waste, lubricant use and, indirectly, energy demand.

The base definition follows the same logic as in flat knitting:

“Input quantity refers to the net yarn required to produce 1 kg of knitted fabric. Yarn losses during the knitting process (thread breaks, edge waste, trial runs) should be included. Typical material losses for knitting range between 1–5%, depending on yarn type and machine efficiency. If no primary data exist, use 2% as default.”

(Laursen et al., 2007; van der Velden et al., 2014; Sandin et al., 2019 – Mistra)

Classification by yarn fineness (dtex)

Yarn category	Tex range	Approx. dtex	Description
Super fine	2 – 7.5 tex	< 75 dtex	Very fine yarns used for lightweight jerseys
Fine	7.5 – 16 tex	75 – 160 dtex	Common in high-quality apparel fabrics
Medium	16 – 40 tex	160 – 400 dtex	Typical for sportswear and medium-weight knits
Coarse	> 40 tex	> 400 dtex	Thicker yarns for heavy or technical fabrics

(Burkinshaw 2016; Banerjee 2014; Ray 2012; Australian Cotton 2022)

Adjustment logic

Parameter	Observation	Representative quote	Source
Yarn loss / waste	Finer yarns are more fragile and break more often during operation.	“Reducing the yarn linear density from 300 to 45 dtex significantly increased yarn breakage during machine operation.”	Haque & Naebe (2024)
Lubricant	More yarn contact and friction increase lubricant demand.	“Yarns with lower dtex required increased lubricant pre-treatment due to higher friction and mechanical interaction.”	Liu, Cano & Ardanuy (2025)
Indirect energy (HVAC, air, etc.)	Lower throughput → more energy per kg.	“Knitting finer yarns resulted in longer machine running time, thereby increasing energy use per kilogram of fabric.”	Palamutcu (2016); Haque & Naebe (2024)
Capital allocation (machine)	Longer knitting time increases depreciation per kg.	“Lower tex yarns required longer processing time, raising capital cost per kg of output.”	INTI (2021)

Parameter values applied

Yarn fineness	Yarn losses (%)	Lubricant (kg/kg fabric)
Super fine (<75 dtex)	5%	0.008
Fine (75–160 dtex)	4%	0.007
Medium (160–400 dtex)	3%	0.006
Coarse (>400 dtex)	2%	0.005

These values are used as a reference to adjust inventory parameters when the yarn fineness is known.

If not, the **medium range** (3% losses, 0.006 kg/kg lubricant) is applied by default.

3. Building Infrastructure, Capital Goods and End-of-Life (EoL)

3.1 Modelling approach

Goal and scope:

The circular knitting process model includes **capital goods (machinery)** and their **end-of-life** as separate mass-based allocations, consistent with EF and ecoinvent guidelines.

We apply the same calculation logic used in the flat knitting process:

$$M_{\text{machine per kg fabric}} = \frac{M_{\text{machine}}}{Q_{\text{lifetime}}}$$

where

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

- **Machine mass** is taken from OEM brochures.
- **Productivity** is taken from manufacturer specifications (e.g., Mayer & Cie., Leadsfon).
- **Service life** parameters vary by scenario (see below), **we are using average.**

3.2 Lifetime parameters (for transparency)

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Resulting allocation (kg/kg fabric)
Best case (BCS)	20	24	365	30	0.0005
Average case	15	16	300	30	0.0013

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Resulting allocation (kg/kg fabric)
Worst case (WCS)	10	8	300	30	0.0039

These values correspond to the **mass of machine attributed per kilogram of fabric**.

They represent the “infrastructure weight” of the process and ensure consistency across product systems.

3.3 End-of-life allocation

Machine composition is assumed **2800kg**:

- **85% steel** → 2380 kg
- **15% other materials** → 450 kg

EuL per kg of fabric (average scenario):

Material	Allocation (kg/kg fabric)	Note
Steel	0.0011	Recycled or disposed depending on scenario
Other materials (Electric and Electronics)	0.0002	Non-ferrous components (motors, electronics)

3.4 Alignment with standards

- Complies with EF guidelines for capital goods inclusion.
- Mirrors ecoinvent’s approach to infrastructure amortisation and decommissioning.
- Neutral and impact-method independent.

3.5 In ecoinvent format

For the “**market for building machine**” dataset:

Field	Entry
Amount	4.6E-07 unit/kg (average scenario)
Unit	unit
Product	building machine
Dataset	market for building machine / GLO

(Derived from 0.0013 kg/kg fabric ÷ 2800 kg machine weight.)

3.6 Building Infrastructure

The circular knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

3.6.1 Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted fabric by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + creel/yarn storage + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

3.6.2 Technology rationale.

Circular knitting is the highest-throughput knitting technology, operating continuously with minimal interruptions. Its compact cylindrical machine footprint combined with high production rates yields the lowest building intensity per kg of output among all knitting technologies considered.

3.6.3 Practical implementation.

In the circular knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 1.00E-05 m²/kg.

3.6.4 End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied:

■ Circular knitting: **1.00E-05 m²/kg**

This value is consistent with the benchmark reported in the openLCA Nexus organic cotton sweater case study (1.61E-05 m²/kg for circular knitting). Physical plausibility was cross-checked using machine envelope data from Terrot GmbH single-jersey machines (e = 2550 mm, k = 2200 mm, envelope ≈ 5.6 m²) with a conservative cell factor of ~6×, yielding $A_{\text{alloc}} \approx$

35 m² and a calculated intensity of ~1.62E-05 m²/kg — same order of magnitude as the adopted default.

4. Indirect Energy Allocation

4.1 Context

For both **flat and circular knitting**, total electricity consumption per kg of fabric is often known, but sub-metered values for subsystems (machines, compressors, HVAC, lighting, cleaning) are rarely available.

To keep models transparent and editable, we allocate total energy into subsystems using percentage factors derived from literature.

4.2 Evidence base

- Factory case showing **80%** of total electricity from HVAC and chillers (Sakti et al., 2021).
- Energy breakdown in spinning: **machines 78%, humidification 16%, lighting 3%, compressors 3%** (Hasanbeigi, 2010).
- Compressed air typically **10–40%** of total plant energy (Confederation of Indian Industry, n.d.).
- Cleaning accounts for **~40% of stoppage time** in knitting (Çelik et al., 2025).
- Weaving process benchmark: **machines 36.3%, compressors 29.4%, HVAC 27.1%**, total **5.06 kWh/kg** (Koç & Çinçik, 2010).

4.3 Default distribution ("mid-range knitting" set)

Subsystem	Share (%)	Comment
Knitting machines	35	Main mechanical load
Compressors	25	Pneumatic system for needle actuation
HVAC	30	Air conditioning and humidification
Lighting	6.5	Facility lighting
Cleaning / auxiliaries	3.5	Maintenance and automation units

This distribution is **identical for flat and circular knitting** until factory-level data allow refinement.

4.4 Total indirect energy

Extrapolating to include HVAC, air compression and auxiliaries, a consistent high-load total of **4.46 kWh/kg** is adopted.

Indirect energy = 4.46 kWh/kg fabric

This figure is used for both **flat and circular knitting** as a temporary reference, pending real measurements.

It can later be split by the above percentages to isolate subsystem consumption.

5. Electricity Consumption

5.1 Concept

Electricity demand in knitting is mainly driven by **yarn fineness (dtex)**, machine speed, and fabric structure.

Finer yarns require longer knitting time and higher frictional work per kilogram of fabric, which increases energy use.

This dependency has been empirically demonstrated by van der Velden et al. (2013) and Sandin et al. (2019), who modelled electricity consumption as inversely proportional to yarn thickness ($E \propto 1/\text{dtex}$).

5.2 Empirical foundation

For knitting, consumption values from Idemat (2012) and ITMF (2010) were interpolated by Sandin et al. (2019), producing the reference equation:

$$E_{\text{circular knitting}} = 0.00303 + \frac{\text{dtex}}{38.805}$$

To improve representativeness, an expanded dataset of **13 anchor points** (80 - 300 dtex) from literature and factory records was used to recalibrate the model.

The updated regression (ordinary least squares) yielded:

$$E_{\text{circular knitting}} = 0.01457 + \frac{37.119}{\text{dtex}} \text{ (kWh/kg fabric)}$$

with **RMSE = 0.0219 kWh/kg** and **R² = 0.967**.

This function remains valid between **80 dtex and 400 dtex**.

Outside this range, values should be cautiously extrapolated! we are use it till 450 dtex and above 80, but is a "low risk".

5.3 Representative values

The model was then applied to generate reference values across typical yarn fineness ranges used in **circular knitting**.

For consistency, the corresponding NE, Nm and denier equivalents were also listed.

Representative Value (kWh/kg)	Circular Knitting dtex	Approx. denier	Nm	NE
0.76	40	45	250	148
0.54	60	63	167	98
0.48	80	72	125	74
0.39	100	90	100	59
0.32	120	108	83	49
0.26	150	135	67	40
0.23	170	153	59	35
0.22	180	162	55	33
0.20	200	180	50	30
0.18	230	207	43	27
0.16	250	225	40	24
0.15	270	245	37	22
0.14	300	270	33	20
0.13	330	297	30	18
0.12	370	333	27	16
0.11	400	470	25	15
0.10	450	420	20	13

(Values rounded; derived from $E=0.01457+37.119/\text{dtex}$)

5.4 Interpretation

- Below 80 dtex, consumption rises sharply as machine efficiency drops.
- For coarser yarns (> 400 dtex), marginal energy reduction is minimal.

5.5 Application in LCI models

When primary data are unavailable, these reference values are used to populate the **electricity input** of knitting unit processes.

The selected value must correspond to the **yarn fineness (dtex)** of the fabric modelled.

For multi-yarn blends, a **mass-weighted dtex** is calculated prior to lookup.

References

- Australian Cotton (2022). *Understanding yarn count and the uses of Australian cotton*.
- Banerjee, P. K. (2014). *Principles of Fabric Formation*. CRC Press.

- Burkinshaw, S. M. (2016). *Physico-chemical Aspects of Textile Colouration*. John Wiley & Sons.
- Çelik, N. et al. (2025). *Design of a washing machine to clean the needle bed of an electronic flat knitting machine*. *JSIR*, 84(1), 17–23.
- Confederation of Indian Industry (n.d.). *Energy Efficiency in Compressed Air Systems*.
- Haque, A. N. M. A., & Naebe, M. (2024). *Future Textiles*. Deakin University.
- Hasanbeigi, A. (2010). *Energy-efficiency improvement opportunities for the textile industry (LBNL-3970E)*.
- INTI (2021). *Product Category Rules: Textile Yarn and Thread*.
- Kır, A., Öztürk, E., Yetiş, Ü., & Kitiş, M. (2024). *Resource utilisation in textile subsectors*. *Environmental Science and Pollution Research*, 31, 25312–25328.
- Koç, E., & Çinçik, E. (2010). *Analysis of energy consumption in woven fabric production*. *Fibres & Textiles in Eastern Europe*, 18(2), 14–20.
- Laursen, S. E. et al. (2007). *EDIPTEx – Environmental Assessment of Textiles*. Danish EPA.
- Liu, Z., Cano, F., & Ardanuy, M. (2025). *Effect of process conditions and lubricant application on yarn quality*. *AUTEX Research Journal*.
- Palamutcu, S. (2016). *Sustainable Textile Technologies*. In *Textiles and Clothing Sustainability*. Springer.
- Ray, S. C. (2012). *Fundamentals and Advances in Knitting Technology*. WPI Publishing.
- Sakti, S. et al. (2021). *Energy efficiency analysis in a textile company using DMAIC approach*. *IOP MSE*, 1096(1), 012007.
- Sandin, G. et al. (2019). *Environmental Assessment of Swedish Fashion Consumption – Mistra Future Fashion*.
- van der Velden, N. M., Patel, M. K., & Vogtländer, J. G. (2014). *LCA Benchmarking Study on Textiles*. *Int. J. LCA*.
- ITMF (2010). *International Production Cost Comparison*.